

# Predicting Benchmark Scores from Unpaired Model Evaluations

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## Abstract

Evaluating a generative model on all available benchmarks is prohibitively expensive. As such, models are often evaluated on different subsets of tasks and/or different subsets of queries from a given task, and the resulting (model  $\times$  task) score matrix is sparse and noisy. In this paper, we introduce the **Product Kernel Perspective Space (PKPS)**—a collection of low-dimensional representations of models based on their embedded responses—that generalizes existing benchmark-prediction methods to the sparse and noisy evaluation regime. Theoretically, we show that PKPS more effectively uses information from unpaired queries for a given task. Empirically, we show that PKPS outperforms sample means, item-response theory, and paired methods where evaluation is sparse, noisy, or unpaired, and that it is a complementary signal to score-based methods.

## 1 Introduction

Generative models are evaluated on an ever-growing collection of benchmarks spanning knowledge, reasoning, code, translation, and safety [Liang et al., 2023]. Evaluating a model on all of them is prohibitively expensive. As a result, models are evaluated on different subsets of tasks and on different subsets of queries within a task: a recent compilation of 84 frontier models on 133 benchmarks finds only 23% of the score matrix filled [Zeng and Papailiopoulos, 2026]. Nor is this an accident of curation—it is how modern evaluation works. Arena-style platforms score each model on whatever conversations its users happen to have [Chiang et al., 2024]; contamination-aware benchmarks refresh their items, so models evaluated in different months answer different questions [White et al., 2024]; and cost-driven subsampling gives each evaluation run its own item subset [Perlitz et al., 2024]. The resulting (model  $\times$  task) score matrix is sparse, its entries are noisy averages over few queries, and the responses beneath it are *unpaired*: two models rarely answer the same queries. Predicting the missing and true entries of this matrix would let practitioners reuse cached evaluations instead of paying to re-run them [Polo et al., 2024, Vivek et al., 2024, Zeng and Papailiopoulos, 2026].

Existing predictors use each cell’s *aggregate* score: low-rank matrix completion imputes missing cells [Zeng and Papailiopoulos, 2026], and item-response theory extrapolates a full score from a scored item subset [Polo et al., 2024]. But the responses that produced those scores carry more information than the scores themselves. The Data Kernel Perspective Space (DKPS) [Helm et al., 2026] uses them: it embeds each model from its query responses via classical multidimensional scaling and regresses benchmark scores in that space, query-efficiently. However, DKPS’s distance between two models is the average difference of their responses *to the same queries*—a *paired* design. On unpaired data it has nothing to compare, and the information in the unpaired responses is thrown away.

In this paper, we introduce the **Product Kernel Perspective Space (PKPS)**, a collection of low-dimensional representations of models based on their embedded responses. PKPS compares responses to *similar*—rather than only identical—queries, so every unpaired response contributes and paired DKPS is recovered as a special case (Section 3). Theoretically, we show that the query budget PKPS requires to reach a target accuracy does not depend on the fraction of queries shared across models (Theorem 1). Empirically, we demonstrate on two heterogeneous suites—93 models on HELM tasks and 45 modern models from the Every Eval Ever datastore [EvalEval Coalition, 2026]—that PKPS achieves lower error than sample means, item-response theory, and DKPS in

Table 1: MAE of benchmark prediction methods on both suites (16 seeds; best per column bold; setup in Section 4.1): query efficiency at budgets  $m \in \{1, 8\}$  queries per cell, and completion of missing cells at the small-cohort, full-cohort ( $p_{\text{task}}=p_{\text{query}}=0.5$ ), and sparse-coverage (sp.:  $p_{\text{task}}=0.2$ ,  $n=10$ ) operating points. “—”: not applicable.

| Method            | HELM suite (18 tasks, 93 models) |              |              |              |              | EEE suite (16 tasks, 45 models) |              |              |              |              |
|-------------------|----------------------------------|--------------|--------------|--------------|--------------|---------------------------------|--------------|--------------|--------------|--------------|
|                   | Query-eff.                       |              | Completion   |              |              | Query-eff.                      |              | Completion   |              |              |
|                   | ( $m=1$ )                        | ( $m=8$ )    | ( $n=10$ )   | ( $n=93$ )   | (sp.)        | ( $m=1$ )                       | ( $m=8$ )    | ( $n=10$ )   | ( $n=45$ )   | (sp.)        |
| sample mean       | 0.292                            | 0.092        | —            | —            | —            | 0.224                           | 0.081        | —            | —            | —            |
| IRT               | 0.348                            | 0.343        | —            | —            | —            | 0.165                           | 0.139        | —            | —            | —            |
| matrix completion | —                                | —            | 0.139        | 0.105        | 0.208        | —                               | —            | 0.109        | 0.096        | 0.144        |
| DKPS              | 0.168                            | 0.109        | —            | —            | —            | 0.116                           | 0.077        | —            | —            | —            |
| PKPS              | 0.136                            | 0.108        | <b>0.120</b> | 0.103        | <b>0.171</b> | <b>0.076</b>                    | 0.058        | <b>0.082</b> | 0.070        | <b>0.108</b> |
| ensemble          | <b>0.132</b>                     | <b>0.077</b> | <b>0.120</b> | <b>0.099</b> | <b>0.171</b> | 0.084                           | <b>0.052</b> | <b>0.082</b> | <b>0.069</b> | <b>0.108</b> |

sparse, noisy, and unpaired regimes, matching the accuracy of sample means with an  $8\times$  smaller query budget, and that ensembling it with score-based methods reduces the error of low-rank completion by up to 28% (Section 4).

## 1.1 Problem setting and notation

A *model* is a random mapping  $f: \mathcal{Q} \rightarrow \mathcal{X}$  from a query space to a response space; querying  $f$  at  $q$  returns a draw from the distribution  $f(q)$ . We observe  $n$  models  $f_1, \dots, f_n$ , each run on its own *query set*  $Q_i \subseteq \mathcal{Q}$ ; the sets may overlap arbitrarily, and the paired design  $Q_1 = \dots = Q_n$  is a special case. Two fixed embedding functions map raw objects to vectors: a *response* embedding  $\text{embed}_R: \mathcal{X} \rightarrow \mathbb{R}^p$  and a *query* embedding  $\text{embed}_Q: \mathcal{Q} \rightarrow \mathbb{R}^{p'}$ . For  $q_j \in Q_i$  we write  $x_{ij} = \frac{1}{r} \sum_{k=1}^r \text{embed}_R(f_i(q_j)^{(k)})$  for the average of the  $r$  sampled response embeddings and  $u_j = \text{embed}_Q(q_j)$  for the embedded query.

Queries are grouped into  $T$  tasks. Model  $i$ ’s *true score* on task  $t$ ,  $y_{it}$ , is its mean score over the task’s full query set; running the model on  $m_{it} \geq 0$  of those queries yields an observed *sample score* whose noise scales as  $1/\sqrt{m_{it}}$ . The goal is to predict  $y_{it}$  for every cell of the (model  $\times$  task) matrix: cells observed with few queries ( $m_{it} > 0$ ; *denoising*, the efficient-benchmarking end) and cells never run ( $m_{it} = 0$ ; *completion*). We write  $y_i$  when the benchmark has a single score. Two properties of the design recur throughout: *coverage*, the fraction of cells with  $m_{it} > 0$ , and *pairing*, the fraction  $\rho$  of a cell’s queries shared across models; real evaluations sit near  $\rho \approx 0$ .

## 2 Related work

**Low-dimensional representations of models.** Some methods compare models through their internals: activations on shared inputs [Duderstadt et al., 2023, Huh et al., 2024, Horwitz et al., 2025] or weights [Chen et al., 2025]. Evaluation, however, is black-box: it sees responses, not internals. The Data Kernel Perspective Space (DKPS) embeds models from their responses via multidimensional scaling [Torgerson, 1952, Helm et al., 2024]; the representations are consistent and concentrate [Acharyya et al., 2024, 2025], support statistical inference [Helm et al., 2025], and are query-efficient [Helm et al., 2026]. But DKPS averages over responses to *shared* queries—a paired design. PKPS keeps the response-level representation and replaces identity matching with a product kernel, so models that answer different queries remain comparable.

**Benchmark prediction.** A second line predicts unobserved (model, task) scores so that not every evaluation must be run; efficient benchmarking and matrix completion are its two ends. Efficient benchmarking scores a small item subset and extrapolates. Item-response theory selects items by difficulty and discrimination [Rasch, 1960, Polo et al., 2024, Kipnis et al., 2024]; anchor points cluster items by cross-model agreement [Vivek et al., 2024]; cognitive-scale embeddings [Bean

et al., 2025] and reinforcement learning [Li et al., 2024] select items adaptively; and design analyses optimize the cost–reliability tradeoff [Perlitz et al., 2024]. Closer to us, DKPS predicts a held-out score from cached responses [Helm et al., 2026], datamodels predict outputs from inputs [Ilyas et al., 2022], and lifelong benchmarks reuse past evaluations [Prabhu et al., 2024]. Matrix completion instead predicts cells that were never run, exploiting low rank across the score matrix [Candès and Recht, 2009, Mazumder et al., 2010, Koren et al., 2009]; BenchPress finds it is approximately rank-two and completes it in logit space [Zeng and Papailiopoulos, 2026]. These methods use only each cell’s *aggregate* score, not its responses. PKPS unifies the two ends—a cell with  $m \geq 0$  queries spans efficient benchmarking ( $m > 0$ ) and completion ( $m = 0$ )—and adds an item-level embedding signal that we ensemble with matrix completion.

### 3 The Product Kernel Perspective Space

**Perspective space.** Each method summarizes the  $n$  models of Section 1.1 in an  $n \times n$  affinity matrix  $A$  (defined below). The squared distance between two models is  $D^2(a, b) = A_{aa} + A_{bb} - 2A_{ab}$  (clamped at 0), and their  $d$ -dimensional *representations* are the classical-multidimensional-scaling minimizer

$$(\hat{\psi}_1, \dots, \hat{\psi}_n) = \arg \min_{z_1, \dots, z_n \in \mathbb{R}^d} \sum_{i,k=1}^n (\|z_i - z_k\| - D(i, k))^2, \quad (1)$$

so model  $i$  is the point  $\hat{\psi}_i \in \mathbb{R}^d$ , its *perspective*. Paired DKPS and PKPS differ only in the affinity  $A$ .

**Paired DKPS.** With all  $n$  models answering the same  $m$  queries ( $Q_1 = \dots = Q_n$ ), DKPS uses the identity-matched affinity  $A_{ab} = \frac{1}{m} \sum_j k_R(x_{aj}, x_{bj})$ : response  $j$  of model  $a$  is paired only with response  $j$  of model  $b$ , so the design must be paired.

**Product kernel.** PKPS replaces identity matching with a query kernel  $k_Q$ :

$$A_{ab} = \frac{\sum_{j \in Q_a} \sum_{l \in Q_b} k_Q(q_j, q_l) k_R(x_{aj}, x_{bl})}{\sum_{j \in Q_a} \sum_{l \in Q_b} k_Q(q_j, q_l)}, \quad (2)$$

where  $Q_a$  is the query set of model  $a$ . Each term is normalized by its own query-kernel mass, so models that answered different numbers of (or entirely different) queries are still comparable; responses to *similar* queries—even across tasks—contribute and unpaired evaluations are used (Figure 1). Paired DKPS is the special case  $k_Q = \delta$ , where only  $j = l$  survives.

**Kernels.** The response kernel  $k_R$  compares embedded responses, the query kernel  $k_Q$  compares embedded queries. Paired DKPS uses a linear response kernel and the identity query kernel  $k_Q = \delta$ . We use an RBF query kernel over the embedded queries,  $k_Q(q_j, q_l) = \exp(-\|u_j - u_l\|^2 / 2\sigma^2)$ , with bandwidth  $\sigma$  chosen per run (below);  $k_R$  is RBF in the synthetic study and linear on real data, where linearity makes the kernel factorize across response domains (Section 4.1). Since Eq. 2 normalizes each pair by its own kernel mass,  $A$  need not be positive semidefinite; classical MDS embeds the doubly-centered matrix by its top- $d$  nonnegative eigenvalues, as is standard.  $A_{ab}$  costs  $|Q_a||Q_b|$  kernel terms per pair, and one response-kernel pass serves the whole bandwidth grid.

#### 3.1 Theoretical properties of the PKPS

PKPS is a strict generalization of DKPS, in two senses. *DKPS limit*: with  $k_Q = \delta$  only equal queries lying in both query sets survive the double sum of Eq. 2, so it reduces to paired DKPS over the shared queries,  $A_{ab} = \frac{1}{|Q_a \cap Q_b|} \sum_{q \in Q_a \cap Q_b} k_R(x_{aq}, x_{bq})$ ; and since the RBF query kernel converges pointwise

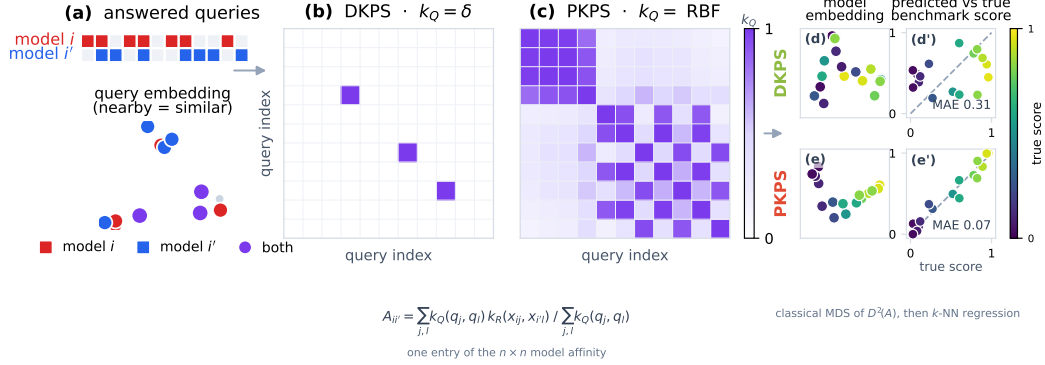


Figure 1: PKPS under *unpaired* evaluation. **(a)** Two models,  $i$  (red) and  $i'$  (blue), answer their own subsets of a query pool (blocks: who answered what; dots: queries in embedding space, nearby = similar). **(b)** DKPS’s identity kernel keeps only the diagonal cells where both models answered the *same* query—almost every response is discarded. **(c)** An RBF  $k_Q$  is a dense similarity matrix: responses to *similar* queries bridge the two models. Both panels index the pooled queries in one shared order; the weights feed one entry of the model affinity (Eq. 2). **(d–e’)** A toy benchmark-prediction example: restricted to the rare shared queries, DKPS’s embedding has no clean score axis and predicts poorly (MAE 0.31); PKPS recovers a smooth score gradient and predicts on the diagonal (MAE 0.07).

to  $\delta$  as  $\sigma \rightarrow 0$  (distinct queries having distinct embeddings), DKPS is also the small-bandwidth limit. *Unpaired comparability*: for any strictly positive  $k_Q$  the normalizer is positive for any pair of nonempty query sets, so  $A_{ab}$  is defined even on disjoint sets, where paired DKPS is undefined. Between the two limits,  $\sigma$  smoothly trades exact matching for cross-query bridging.

### 3.2 Inference in the PKPS

Inference is a two-step statistical problem, exactly as in the DKPS [Helm et al., 2026]: embed each model into the perspective space by the scaling of Eq. 1, then regress its benchmark score on the resulting coordinate. Concretely, on a set of reference models with known scores  $\{(f_i, y_i)\}$  we fit a decision function  $\hat{h} : \mathbb{R}^d \rightarrow \mathcal{Y}$  on the pairs  $\{(\hat{\psi}_i, y_i)\}$  and predict any model’s score as  $\hat{y}(f) = \hat{h}(\hat{\psi}(f))$ . The only free choices are the query-kernel bandwidth  $\sigma$  and the regressor  $\hat{h}$ , which we describe next; we then add the score channel that turns the embedding into an ensemble.

**Choosing the bandwidth.** The optimal  $\sigma$  depends on how paired the data is: small when query overlap is high (exact matching), large when it is low (bridging). We set it per run by cross-validation over a median-scaled grid  $\sigma \in \{0.03, \dots, 3\} \cdot \text{med}$  (median pairwise query distance), selecting the  $\sigma$  whose perspective best predicts the *observed* scores under leave-one-model-out  $k$ -NN—never the held-out scores we report, so the choice is not circular. Because the grid reaches near-delta, cross-validation *recovers* DKPS when overlap is high and bridges only when it must.

**Estimation pipeline.** For each replicate we (i) PCA-reduce response and query embeddings on the *observed* rows only (dimension by the second Zhu–Ghodsí elbow); (ii) form  $D$  from Eq. 2; (iii) embed models by classical MDS; (iv) map the embedding to scores with the *same* head as the matrix-completion baseline up to the factor model—logit transform, per-task standardization, two-way (model + task) bias—then  $k$ -NN regression of the residual on the embedding, fit leave-one-family-out and predicting every cell. Matching the per-task standardization matters: without it the model offset is dominated by the highest-logit-variance tasks, biasing the embedding on multi-scale suites.

**The embedding- $\oplus$ -score ensemble.** Each cell combines the embedding with whichever score signal carries independent information. On a *missing* cell that is matrix completion [Zeng and Papailiopoulos, 2026], blended with PKPS by a weight fit by held-out cross-validation on observed scores. On an *observed* cell it is the cell’s own sample mean, shrunk toward the PKPS estimate by the precision weight  $\sigma_q^2/(\sigma_q^2 + e_{PKPS}^2)$ —sample noise variance over total disagreement with PKPS—so the sample dominates as queries grow and the embedding as noise grows. Both quantities are read off observed queries ( $\hat{p}(1 - \hat{p})/m$ ; depth-weighted disagreement), so the weight never touches held-out scores and the score baselines see exactly the same information.

### 3.3 Query efficiency

PKPS is designed to reach a target prediction accuracy from fewer evaluated queries than paired DKPS: by bridging responses to similar queries it uses evidence that DKPS, limited to shared queries, discards. We formalize this by extending the machinery of Helm et al. [2026].

**Definition 1** (Population PKPS). Let each model  $a$ ’s (query, response) pairs be drawn i.i.d. from a design distribution  $P_a$ . The *population affinity* is  $A_{ab}^* = \mathbb{E}[k_Q(u, u') k_R(x_a, x_b)] / \mathbb{E}[k_Q(u, u')]$  with draws from  $P_a$  and  $P_b$  independent, the population squared distances are  $D^{*2}(a, b) = A_{aa}^* + A_{bb}^* - 2A_{ab}^*$ , and the *population perspectives*  $\psi_1, \dots, \psi_n \in \mathbb{R}^d$  are the rank- $d$  spectral embedding of the centered Gram matrix  $B^*$  of  $D^{*2}$ .

**Assumption 1** (Score regularity). The benchmark score is bounded and  $\gamma$ -Lipschitz on the population perspective space.

**Assumption 2** (Model support). The model distribution places mass on every neighborhood of the population perspective space.

**Assumption 3** (Kernel and geometry regularity). Kernels and embedded responses are bounded; the *cross-model query-kernel mass* is bounded below,  $\kappa = \inf_{a,b} \mathbb{E}[k_Q(u, u')] > 0$ ; and  $B^*$  has rank  $d$  with spectral gap  $\lambda_d > 0$ . ( $\kappa$  is the population analogue of  $\rho$ : positive for an RBF  $k_Q$  whenever query distributions overlap, even with no shared query.)

**Proposition 1** (Concentration of PKPS perspectives). *Under Assumption 3, with at least  $m$  queries per model, for any  $\delta \in (0, 1)$ , with probability at least  $1 - \delta$  there is an orthogonal  $W$  such that  $\max_i \|\hat{\psi}_i - W \psi_i\| \leq C \kappa^{-2} \lambda_d^{-1/2} \sqrt{d} n \sqrt{\log(4n^2/\delta)/2m}$ , where  $C$  depends only on the kernel bounds (proof in Appendix D).*

**Definition 2** (Query efficiency; following Helm et al., 2026). For a score estimator  $h$  and a query design with per-model budget  $m$ , let  $m_h(\epsilon)$  be the smallest budget at which  $h$  attains  $\text{MSE} \leq \epsilon$  with high probability for some number of reference models. Estimator  $h$  is *query-efficient relative to  $h'$*  if  $m_h(\epsilon) \leq m_{h'}(\epsilon)$  for all sufficiently small  $\epsilon$ .

**Theorem 1** (Query efficiency of PKPS). *Under Assumptions 1–3, for any  $\epsilon > 0$  there exist  $(n, m)$ , with  $m$  polynomial in  $(1/\epsilon, 1/\kappa, n)$  and independent of the pairing fraction  $\rho$ , such that nearest-neighbor regression in the PKPS attains mean squared error at most  $\epsilon$  with high probability; in particular  $m_{PKPS}(\epsilon)$  does not grow as  $\rho \rightarrow 0$ .*

*Proof sketch.* Proposition 1 bounds the estimation error of every perspective. By Assumption 1 and the triangle inequality, the nearest-neighbor regression error splits into two estimation terms and the population nearest-neighbor distance, which Assumption 2 makes small for  $n$  large; choosing  $m = \text{poly}(n, 1/\kappa, 1/\epsilon)$  gives  $\text{MSE} \leq \epsilon$ , and no step involves  $\rho$ . Full proof in Appendix D.  $\square$

Applied to paired DKPS ( $k_Q = \delta$ ), the same argument retains only the  $\rho M_{ij}$  shared query pairs, so the DKPS bound’s effective budget is  $\rho m$  and its guarantee degrades as  $\rho \rightarrow 0$  while Theorem 1 does not. In the sense of Definition 2 we conjecture strict separation; Section 4.2 is its empirical counterpart.

**Conjecture 1** (Strict gap). *Under Assumptions 1–3, PKPS is query-efficient relative to paired DKPS, with  $m_{\text{DKPS}}(\epsilon)/m_{\text{PKPS}}(\epsilon)$  increasing as  $\rho \rightarrow 0$ ; the two coincide in the fully paired limit  $\rho = 1$ .*

## 4 Experiments

We evaluate PKPS on synthetic data, where every lever of the observation process can be swept, and on two heterogeneous benchmark suites. Throughout, the two prediction problems of Section 1.1 structure the experiments: denoising observed cells (query efficiency) and predicting missing ones (completion).

### 4.1 Evaluation setup

**Synthetic data.** We generate  $n$  models with latent ability vectors  $v_i$  and  $T$  tasks with latent vectors  $t_k$ , scores  $v_i \cdot t_k + \epsilon$ , and queries drawn near each task center. Each model answers  $M_{ij}$  queries per task, of which  $m_{ii'}$  are shared with another model (overlap  $\rho = m_{ii'}/M_{ij}$ ); a task is observed with probability  $p_{\text{task}}$  and a query with probability  $p_{\text{query}}$ .

**Benchmark suites.** The first suite is 18 tasks across four HELM datasets with very different response types—MATH (7 math subjects), WMT-14 (5 translation directions), MedQA (medical multiple-choice), and LegalBench (5 legal classification subsets)—on the 93 models evaluated on all four. MATH and WMT have rich free-text responses; MedQA and LegalBench responses are single tokens for which text embeddings are degenerate, so we encode each dataset’s responses natively (Gemini embeddings vs. one-hot answers) in disjoint blocks. With a linear  $k_R$  this makes cross-dataset response similarity exactly zero, and a single shared query embedding with a within-domain bandwidth keeps  $k_Q$  near zero across domains: the product kernel factorizes by domain, never comparing a proof to a legal label. The second suite, from the Every Eval Ever datastore [EvalEval Coalition, 2026], is 45 modern models on 16 tasks (MATH-MC levels 1–5, GSM-MC, GPQA-Diamond, and nine judging categories from JudgeBench and RewardBench 2), with per-cell query pools drawn independently per model ( $\sim 10\%$  incidental overlap)—unpaired by construction (Appendix B).

**Prediction methods.** We compare six predictors. The *sample mean* is a cell’s own  $m$ -query average. *IRT* fits a one-parameter logistic model on the largest shared item block [Polo et al., 2024]. *Matrix completion* is the logit-space low-rank imputation of Zeng and Papailiopoulos [2026] (missing cells only). *DKPS* and *PKPS* regress scores on the model embeddings of Section 3, DKPS being the identity-query-kernel limit. The *ensemble* combines PKPS with whichever score signal a cell has (Section 3).

**Measurement.** Every experiment predicts held-out or true cell scores and reports MAE against the full-evaluation score, mean  $\pm 1$  SEM over 16 seeds (50 on synthetic data). Sweeps vary one observation lever at a time—queries per cell  $m$  (noise), task coverage  $p_{\text{task}}$ , cohort size  $n$ , and, on synthetic data, the pairing  $\rho$ —holding the others fixed (grids in Appendix A); all hyperparameters, including the query bandwidth, are chosen per run by the leak-free cross-validation of Section 3.

### 4.2 Results

**Synthetic study.** We predict held-out (model, task) scores, comparing PKPS (RBF query kernel) against DKPS (identity) with the score-only sample baseline as a floor (Figure 2). PKPS—selecting



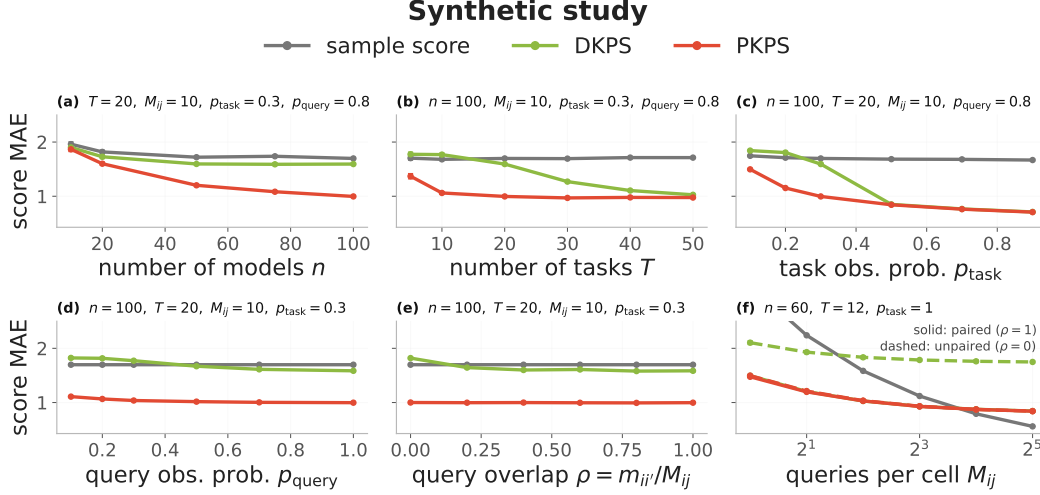


Figure 2: Synthetic study (score MAE,  $\pm 1$  SEM over 50 seeds). DKPS (identity query kernel, green) vs. PKPS (RBF, red), with the score-only sample baseline (gray); each panel sweeps one parameter with the rest fixed (shown in its title). (a)–(e) **complete held-out cells** from observed responses—cohort  $n$ , tasks  $T$ , task coverage  $p_{\text{task}}$ , query coverage  $p_{\text{query}}$ , and cross-model overlap  $\rho = m_{ij'}/M_{ij}$ . PKPS (CV-selected bandwidth) beats DKPS throughout—by a wide margin even when mostly paired (a–c), since it bridges across tasks as well as within—and the gap is largest when unpaired or sparse (d, e), where in (e) DKPS collapses toward the floor as  $\rho \rightarrow 0$  while PKPS stays flat. (f) **denoises observed cells**: each cell is measured with  $M_{ij}$  noisy queries and we predict its true score, at paired ( $\rho=1$ , solid) and unpaired ( $\rho=0$ , dashed). The raw sample falls  $\propto 1/\sqrt{M_{ij}}$ ; smoothing over the perspective denoises well below it at small budgets—PKPS paired or unpaired, DKPS only paired—until the sample becomes precise at the full budget.

its bandwidth by the same leak-free cross-validation as on real data—beats DKPS on *every* lever, and both beat the sample floor: bridging similar queries (across tasks as well as within) supplies signal exact-match DKPS cannot reach, so the margin is wide even when mostly paired (a–c). The gap peaks where pairing is scarce: as  $p_{\text{query}}$  drops (d) DKPS degrades while PKPS stays flat, and sweeping  $\rho$  directly (e), DKPS collapses toward the floor as  $\rho \rightarrow 0$  while PKPS is flat—it does not rely on pairing.

Panel (f) is the *denoising* form of the same problem: each cell is now *observed* with  $M_{ij}$  noisy queries (sample noise  $\propto 1/\sqrt{M_{ij}}$ ) and we predict its true score. The raw sample mean (gray) falls like  $1/\sqrt{M_{ij}}$ ; smoothing those noisy scores over the perspective denoises well below it while measurement noise dominates—PKPS reaches at a single query the accuracy the raw sample needs roughly five queries to match, and does so whether queries are paired or not, while DKPS denoises only when paired (unpaired DKPS, dashed, barely improves). Only once the budget is large is the raw sample precise enough to overtake the embedding’s residual bias. This is the empirical counterpart of Theorem 1 and Conjecture 1.

**Query efficiency.** Both real-data applications are inherently *unpaired*—each model is run on its own queries, the real-data counterpart of the  $\rho \rightarrow 0$  limit above. Because the embedding captures response *style*, which is stable even when the per-cell sample mean is noisy, it denoises cheap evaluations. Each model answers  $m$  queries per cell, sampled independently—its own unpaired regime, on which IRT is poorly identified; we predict the true full-eval score (Figure 3). At a small budget the embedding is far more accurate than the raw mean: at  $m=1$ , PKPS scores 0.136 and the ensemble 0.132 versus the sample’s 0.292; PKPS at one query matches the sample at four. As the budget grows the raw mean catches up and eventually overtakes the embedding’s residual bias (0.034 vs. 0.085 at  $m=32$ ); the leak-free ensemble tracks the better channel to within a few thousandths. Panels

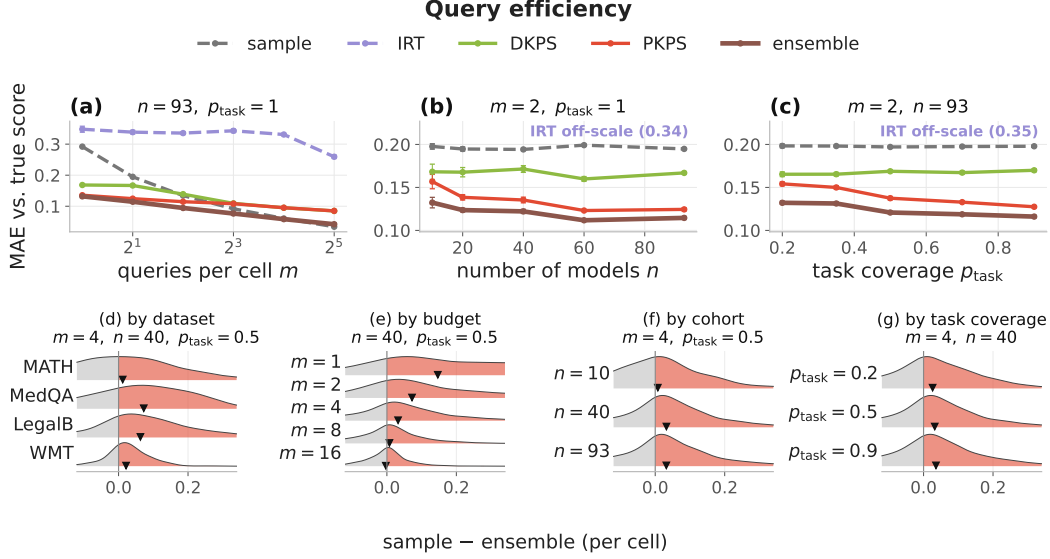


Figure 3: **Query efficiency.** (a–c) Each model answers  $m$  unpaired queries per *observed* cell; predict its true full-eval score (MAE,  $\pm 1$  SEM). PKPS (red) and the ensemble (brown) denoise far below the raw sample (grey) at small budgets and across cohort (b) and coverage (c); DKPS (green) trails on unpaired queries, and IRT (purple) is poorly identified without a shared item block (omitted from b, c; value annotated). (d–g) Conditional breakdown: per-cell  $\Delta = \text{sample} - \text{ensemble}$  MAE ( $\Delta > 0$ , red = win) by dataset, budget, cohort, and coverage, other levers at their sweep median. Broad but tail-driven; the per-row mean ( $\blacktriangledown$ ) is positive throughout.

(b, c) separate the embeddings: PKPS improves steadily with cohort and coverage ( $0.157 \rightarrow 0.124$ ,  $0.154 \rightarrow 0.128$ ; 35–37% on the second suite) while DKPS stays flat—more unpaired data helps only the method that can use it. Per cell, the denoising win is broad but tail-driven: the ensemble beats the raw sample on most cells, with the largest gains at the smallest budgets and on the rich-response datasets (Figure 3d–g).

**Matrix completion.** The same embedding also predicts cells that were *never run*. We observe a fraction  $p_{\text{task}}$  of cells (each at depth  $p_{\text{query}}$ ) and predict the missing ones, against a low-rank matrix-completion baseline (the sample does not apply—a missing cell has no sample). The contrast is the cohort: low-rank completion needs many rows, so when models are scarce it fails for lack of them while the embedding, which needs none, remains accurate—at  $n=10$ , completion is 0.139 versus PKPS’s 0.120 (Figure 4, dashed). At full cohort ( $n=93$ , solid) PKPS slightly edges completion (0.103 vs. 0.105) and the ensemble is best (0.099). Even in the regime most favorable to it—a dense, low-rank matrix—completion is matched or slightly beaten; the embedding’s clear advantage is the sparse-cohort regime completion cannot reach. The per-cell win over completion is modest and tail-driven—completion is strong on this low-rank suite—but stays positive in the mean across datasets, cohorts, coverage, and query depth (Figure 4e–h).

**Second suite.** Both applications replicate on the Every Eval Ever suite and sharpen the completion contrast (Table 1; lever sweeps in Appendix B): at  $m=1$  PKPS attains a third of the sample’s error (0.076 vs. 0.224), and on missing cells the ensemble beats completion at *every* operating point, including full cohort (0.069 vs. 0.096 at  $n=45$ )—this matrix has higher effective rank, so the response channel wins even where completion is strongest. Table 1 collects the two applications on both suites: PKPS is competitive in every column, and the ensemble, which folds in whichever score signal exists, is best or tied nearly everywhere.



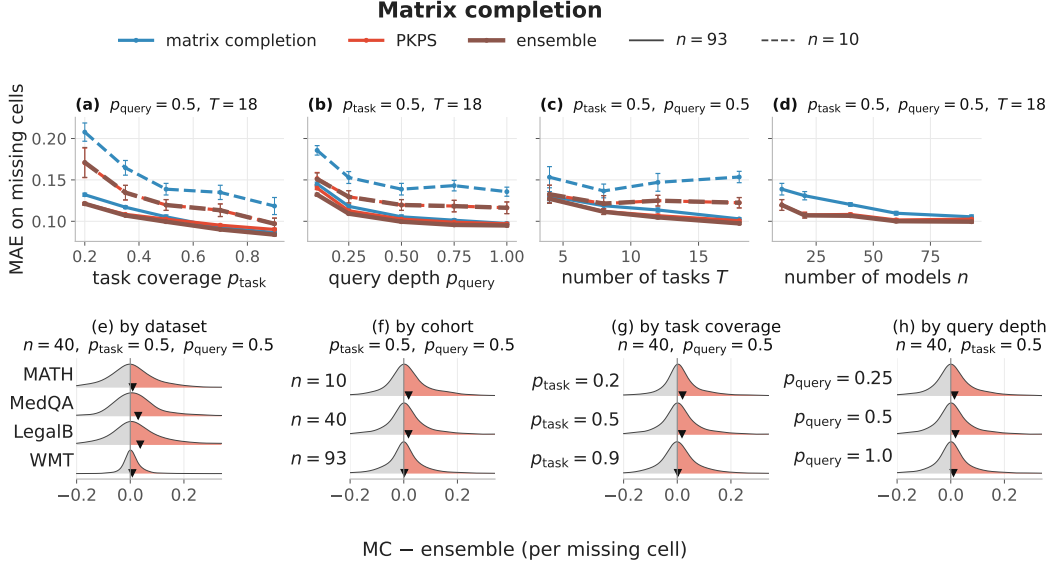


Figure 4: **Matrix completion.** (a–d) Observe a fraction of (model, task) cells, predict the *missing* ones (MAE on missing cells,  $\pm 1$  SEM). Colour is method (low-rank completion blue, PKPS red, ensemble brown); line style is cohort (solid  $n=93$ , dashed  $n=10$ ). The embedding rescues the small-cohort regime where low-rank completion collapses for lack of rows, and ties it at full cohort. (e–h) Conditional breakdown: per-missing-cell  $\Delta = \text{completion} - \text{ensemble}$  MAE ( $\Delta > 0$ , red = win) by dataset, cohort, coverage, and query depth, other levers at their sweep median. Modest and tail-driven on this low-rank suite, but the per-row mean ( $\blacktriangledown$ ) stays positive.

## 5 Discussion

Real leaderboards are organic: each model answers whatever queries its authors chose, so the responses benchmark estimation must compare are *unpaired*—and methods that need a shared block of identical queries collapse there. PKPS removes that requirement with a product kernel comparing responses to *similar* queries, recovering paired DKPS in its  $\sigma \rightarrow 0$  limit. Our central finding is that benchmark estimation survives unpaired evaluation: as  $\rho \rightarrow 0$ , DKPS and IRT degrade sharply while PKPS holds. That capability yields two applications from one embedding—denoising cheap, few-query evaluations and predicting never-run cells—and the ensemble with a cell’s own sample is never the worse choice. None of this assumes pairing or exhaustive evaluation.

**Limitations.** PKPS embeds every observed response and query; cost and quality inherit from the embedders (not ablated). Our block-diagonal instantiation disables cross-domain bridging; a shared response space is untested. The affinity is quadratic in queries per model pair (approximation needed only for very large logs). Theorem 1 is existential (Assumption 2 gives no rate in  $n$ ); the strict DKPS separation remains a conjecture. Finally, per-cell wins over score-only baselines are modest on dense low-rank matrices and decisive in the sparse, noisy, unpaired regime that motivates the problem.

### Reproducibility statement

All methods are released as a documented Python package (the PKPS estimator, matrix completion, and IRT baselines), with the data-preparation, experiment, and figure scripts for both the synthetic study and the HELM suite; every experiment is seeded, all real-data numbers are means over 16 seeds (synthetic: 50), and hyperparameters are selected by the leak-free cross-validation described in Section 3 (leakage is checked by an automated test that corrupts one model’s held-out scores and

verifies its predictions are unchanged). Appendix A lists the suite composition, embedding models, and all hyperparameter settings.

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## A Experimental details

**Suite.** 18 tasks over four datasets from HELM [Liang et al., 2023]: MATH (7 subjects), WMT-14 (5 translation directions), MedQA, and LegalBench (5 subsets), on the 93 models evaluated on all four; Table 2 lists every task with its query count (the loader subsamples each task to at most 120 queries with a fixed seed, so datasets contribute comparably). MATH and WMT-14 responses are free text and are embedded with `gemini-embedding-001` (3072-d); MedQA and LegalBench responses are single-token answers, encoded one-hot. Queries share a single `gemini-embedding-001` embedding space. Response and query embeddings are reduced by PCA fit on observed rows only, with dimension chosen by the second Zhu–Ghodsí elbow.

Table 2: HELM suite composition: 18 tasks over four datasets, with the number of unique queries per task. Every task is scored for all 93 shared models; the loader caps each task at 120 queries (seeded subsample).

| Dataset    | Task                         | Queries | Dataset    | Task                  | Queries |
|------------|------------------------------|---------|------------|-----------------------|---------|
| MATH       | algebra                      | 135     | WMT-14     | cs-en                 | 873     |
| MATH       | counting & probability       | 39      | WMT-14     | de-en                 | 776     |
| MATH       | geometry                     | 38      | WMT-14     | fr-en                 | 754     |
| MATH       | intermediate algebra         | 52      | WMT-14     | hi-en                 | 387     |
| MATH       | number theory                | 30      | WMT-14     | ru-en                 | 613     |
| MATH       | prealgebra                   | 86      | MedQA      | med.qa                | 1000    |
| MATH       | precalculus                  | 57      | LegalBench | abercrombie           | 95      |
| LegalBench | corporate lobbying           | 490     | LegalBench | citizenship questions | 1000    |
| LegalBench | function of decision section | 367     | LegalBench | proa                  | 95      |

**Model coverage (HELM).** The 93 shared models span 20 developers: 01-ai-yi-34b, 01-ai-yi-6b, 01-ai-yi-large-preview, AlephAlpha.luminous-base, AlephAlpha.luminous-extended, AlephAlpha.luminous-supreme, ai21-j2-grande, ai21-j2-jumbo, ai21-jamba-1.5-large, ai21-jamba-1.5-mini, ai21-jamba-instruct, allenai.olmo-7b, amazon.nova-lite-v1, amazon.nova-micro-v1, amazon.nova-pro-v1, anthropic.claude-2.0, anthropic.claude-2.1, anthropic.claude-3-5-haiku-20241022, anthropic.claude-3-5-sonnet-20240620, anthropic.claude-3-5-sonnet-20241022, anthropic.claude-3-haiku-20240307, anthropic.claude-3-opus-20240229, anthropic.claude-3-sonnet-20240229, anthropic.claude-instant-1.2, anthropic.claude-instant-v1, anthropic.claude-v1.3, cohere.command, cohere.command-light, cohere.command-r, cohere.command-r-plus, databricks.dbrx-instruct, deepseek-ai-deepseek-llm-67b-chat, deepseek-ai-deepseek-v3, google.gemini-1.0-pro-001, google.gemini-1.0-pro-002, google.gemini-1.5-flash-001, google.gemini-1.5-flash-002, google.gemini-1.5-pro-001, google.gemini-1.5-pro-002, google.gemini-1.5-pro-preview-0409, google.gemini-2.0-flash-exp, google.gemma-2-27b-it, google.gemma-2-9b-it, google.gemma-7b, google.text-bison@001, google.text-unicorn@001, meta.llama-2-13b, meta.llama-2-70b, meta.llama-2-7b, meta.llama-3-70b, meta.llama-3-8b, meta.llama-3.1-405b-instruct-turbo, meta.llama-3.1-70b-instruct-turbo, meta.llama-3.1-8b-instruct-turbo, meta.llama-3.2-11b-vision-instruct-turbo, meta.llama-3.2-90b-vision-instruct-turbo, meta.llama-3.3-70b-instruct-turbo, meta.llama-65b, microsoft.phi-2, microsoft.phi-3-medium-4k-instruct, microsoft.phi-3-small-8k-instruct, mistralai.mistral-7b-instruct-v0.3, mistralai.mistral-7b-v0.1, mistralai.mistral-large-2407, mistralai.mistral-medium-2312, mistralai.mixtral-8x22b, mistralai.mixtral-8x7b-32kseqn, mistralai.open-mistral-nemo-2407, nvidia.nemotron-4-340b-instruct, openai.gpt-3.5-turbo-0613, openai.gpt-4-0613, openai.gpt-4-1106-preview, openai.gpt-4-turbo-2024-04-09, openai.gpt-4o-2024-05-13, openai.gpt-4o-2024-08-06, openai.gpt-4o-mini-2024-07-18, openai.text-davinci-002, openai.text-davinci-003, qwen.qwen1.5-110b-chat, qwen.qwen1.5-14b, qwen.qwen1.5-32b, qwen.qwen1.5-72b, qwen.qwen1.5-7b, qwen.qwen2-72b-instruct, qwen.qwen2.5-72b-instruct-turbo, qwen.qwen2.5-7b-instruct-turbo, snowflake.snowflake-arctic-instruct, tiuuai.falcon-40b, tiuuai.falcon-7b, upstage.solar-pro-241126, writer.palmyra-x-004, writer.palmyra-x-v2, writer.palmyra-x-v3.

**PKPS.** RBF query kernel with bandwidth selected per run by leave-one-model-out  $k$ -NN ( $k=8$ ) on the observed scores over the grid  $\sigma \in \{0.03, 0.1, 0.3, 1, 3\} \times$  median pairwise query distance, plus the near-delta (DKPS) limit; linear response kernel on real data, RBF on synthetic. Response blocks are capped at 48 PCA components per dataset and the shared query space at 64; each (model, query) pair uses one sampled response ( $r=1$ ). Models are embedded by classical MDS with  $d=8$  throughout, following Helm et al. [2026] and fixed a priori ( $d$  = latent dimension on synthetic). Appendix C shows the results are insensitive to  $d$  over 4–24, and that the cross-validated bandwidth tracks the best fixed choice.

**Prediction head.** All embedding methods share one head: logit-transformed scores, per-task standardization to unit variance, a two-way (model + task) additive bias, and a distance-weighted  $k$ -NN ( $k=5$ ) regression of the residual on the model embedding, fit *leave-one-family-out*: all models from the target’s developer are excluded from its training set, so predictions never lean on near-duplicate siblings [Helm et al., 2026]. Held-out cells are predicted directly; observed cells leave their own target out.

**Baselines.** Matrix completion follows BenchPress [Zeng and Papailiopoulos, 2026]: logit space, per-task standardization, two-way bias, rank-2 ALS with ridge  $\lambda=0.1$ , averaged over 5 random initializations; on observed cells it is cross-fit so a cell’s prediction never sees its own sample. IRT is a 1PL (Rasch) model on binary tasks: item difficulties by maximum likelihood (L-BFGS) on the largest fully-observed (model  $\times$  item) block, per-model ability by bounded scalar MLE, clipped to  $[-4, 4]$ . The sample baseline is the cell’s own  $m$ -query mean.

**Ensemble.** On observed cells, the sample mean is shrunk toward the PKPS estimate with per-cell precision weight  $e_{PKPS}^2 / (e_{PKPS}^2 + \hat{p}(1 - \hat{p})/m)$  on the sample, where  $e_{PKPS}^2$  is the embedding’s depth-weighted disagreement with the observed samples of that model; on missing cells, PKPS and matrix completion are blended with a weight fit by held-out cross-validation against observed scores. No quantity ever uses the held-out full-eval scores.

**Sweeps and reporting.** All real-data sweeps use 16 seeds; curves and table entries report the per-seed mean over tasks, then the mean ( $\pm 1$  SEM) over seeds. Query efficiency: budgets  $m \in \{1, 2, 4, 8, 16, 32\}$  at full cohort and coverage; cohort  $n \in \{10, 20, 40, 60, 93\}$  (HELM) or  $\{10, 20, 30, 45\}$  (EEE) and task coverage  $p_{\text{task}} \in \{0.2, 0.35, 0.5, 0.7, 0.9\}$ , both at the few-query point  $m=2$ . Completion: task coverage as above and query depth  $p_{\text{query}} \in \{0.1, 0.25, 0.5, 0.75, 1.0\}$  with cohort lines  $n \in \{10, \text{full}\}$ ; task count  $T \in \{4, 8, 12, 18\}$  (HELM) or  $\{4, 8, 12, 16\}$  (EEE); the cohort sweep holds  $p_{\text{task}}=p_{\text{query}}=0.5$ .

**Synthetic study.** 50 seeds per configuration; latent dimension 5, observed embedding dimension 20, score noise 0.1, response noise 0.5; both kernels RBF;  $k$ -NN regression with  $k=5$ . Panel-specific settings are given in each figure title.

**Compute.** All experiments run on a single multi-core CPU node; text embeddings are precomputed once through the embedding API and cached. Embedding the EEE suite (30,304 pooled responses plus 7,665 queries,  $\sim 11\text{M}$  tokens) took  $\sim 13$  minutes and under \$2 of API credit; each 16-seed sweep completes in minutes (EEE) to hours (HELM) on the same node.

## B The Every Eval Ever suite

The second suite is built from the Every Eval Ever datastore [EvalEval Coalition, 2026], an MIT-licensed, community-maintained collection of item-level evaluation results in a unified format (inputs, raw model outputs, and per-item scores). We take the five benchmarks with the widest item-level model coverage—MATH-MC, GSM-MC, GPQA-Diamond, JudgeBench, and RewardBench 2—restricted to the 45 models present on all five; sub-tasks (MATH-MC levels 1–5; four JudgeBench and five RewardBench 2 categories) give 16 tasks (Table 3). Every domain is free text (including the judging benchmarks’ rationales), so all responses and queries are embedded with `gemini-embedding-001` and blocked by benchmark exactly as in Appendix A. Each cell’s observable pool is 32 queries drawn with a model-dependent seed, so two models share only incidental queries ( $\sim 10\%$ ); the query-efficiency budget  $m$  and the completion depth  $p_{\text{query}}$  subsample this pool, and the prediction target is always the cell’s full-depth mean over all items, which is never embedded. All scores are binary, so IRT applies to every task (on the HELM suite it excludes WMT). Figures 5 and 6 repeat the lever sweeps of Figures 3 and 4 on this suite; Table 1 summarizes.

**Model coverage (EEE).** The 45 models shared by all five benchmarks span 13 developers: `cohere/c4ai-command-a-03-2025`, `cohere/c4ai-command-r-08-2024`, `cohere/c4ai-command-r-plus-08-2024`, `cohere/c4ai-command-r7b-12-2024`, `cohere/command-a-reasoning-08-2025`, `deepseek/deepseek-r1-0528`,

Table 3: EEE suite composition: 16 tasks over five benchmarks, with the number of items per (model, task) cell. Every task is fully answered by all 45 shared models; each cell’s observable pool is 32 items, sampled per model.

| Benchmark     | Task         | Items | Benchmark     | Task       | Items |
|---------------|--------------|-------|---------------|------------|-------|
| MATH-MC       | level 1      | 430   | JudgeBench    | coding     | 42    |
| MATH-MC       | level 2      | 882   | JudgeBench    | knowledge  | 154   |
| MATH-MC       | level 3      | 1115  | JudgeBench    | math       | 56    |
| MATH-MC       | level 4      | 1199  | JudgeBench    | reasoning  | 98    |
| MATH-MC       | level 5      | 1288  | RewardBench 2 | factuality | 475   |
| GSM-MC        | gsm_mc       | 1319  | RewardBench 2 | focus      | 495   |
| GPQA-Diamond  | gpqa.diamond | 198   | RewardBench 2 | math       | 183   |
| RewardBench 2 | precise if   | 160   | RewardBench 2 | safety     | 450   |

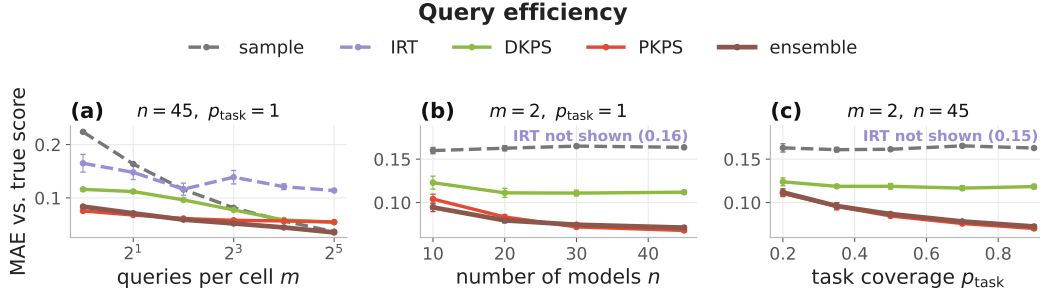


Figure 5: **Query efficiency on the Every Eval Ever suite** (45 models, 16 tasks; MAE vs. full-depth score,  $\pm 1$  SEM over 16 seeds). Same protocol as Figure 3. PKPS at  $m=1$  attains a third of the sample’s error and the sample needs  $m \approx 8$  to catch up; DKPS trails because pools are unpaired, and IRT is poorly identified. The ensemble is best from  $m \geq 2$  on.

deepseek/deepseek-v3-1-terminus, deepseek/deepseek-v3-2, deepseek/deepseek-v3-2-speciale, deepseek/deepseek-v4-flash-fp8, deepseek/deepseek-v4-pro, google/gemini-3-1-pro-preview, google/gemma-2-27b-it, google/gemma-2-9b-it, google/gemma-3-12b-it, google/gemma-3-27b-it, google/gemma-4-e2b-it, google/gemma-4-e4b-it, llm360/k2-v2-instruct, meta/llama-4-maverick-17b-128e-instruct-fp8, meta/meta-llama-3-1-8b-instruct, minimax/minimax-m2-5, mistral/mistral-small-4-119b-2603, moonshot/kimi-k2-5, moonshot/kimi-k2-6, moonshot/kimi-k2-thinking, openai/gpt-5-5, openai/gpt-5-mini, openai/gpt-5-nano, openai/gpt-oss-120b, openai/gpt-oss-20b, qwen/qwen3-30b-a3b-thinking-2507, qwen/qwen3-5-122b-a10b, qwen/qwen3-5-2b, qwen/qwen3-5-397b-a17b, qwen/qwen3-6-35b-a3b, qwen/qwen3-6-plus, qwen/qwen3-next-80b-a3b-thinking, stepfun/step-3-5-flash, xiaomi/mimo-v2-flash, zai-org/glm-4-6-fp8, zai-org/glm-4-7-flash, zai-org/glm-4-7-fp8, zai-org/glm-5-1-fp8, zai-org/glm-5-fp8.

## C Sensitivity analyses

Figure 7 probes the two representation hyperparameters on the query-efficiency experiment (16 seeds throughout). **MDS dimension (a, b)**. Repeating the budget sweep at  $d \in \{4, 8, 24\}$  moves PKPS’s MAE by less than 0.005 at every budget on both suites (at the smallest budgets, smaller  $d$  is very slightly better—fewer coordinates to estimate from one noisy query per cell); the paper’s choice ( $d=8$ , following Helm et al., 2026) is never the worst point at any budget. **Query bandwidth (c)**. Fixing the bandwidth at each grid multiplier instead of cross-validating traces the expected curve: near-delta multipliers and the exact-matching  $\delta$  limit are substantially worse (unpaired pools leave them little to compare), broad multipliers are best, and the leak-free cross-validated choice matches the best fixed multiplier on both suites (0.126 vs. 0.125 on HELM, 0.069 vs. 0.068 on EEE)—the one hyperparameter that matters is the one the method selects automatically. Sensitivity to the number of reference models is part of the main experiments (the cohort sweeps of Figures 3b and 4d).



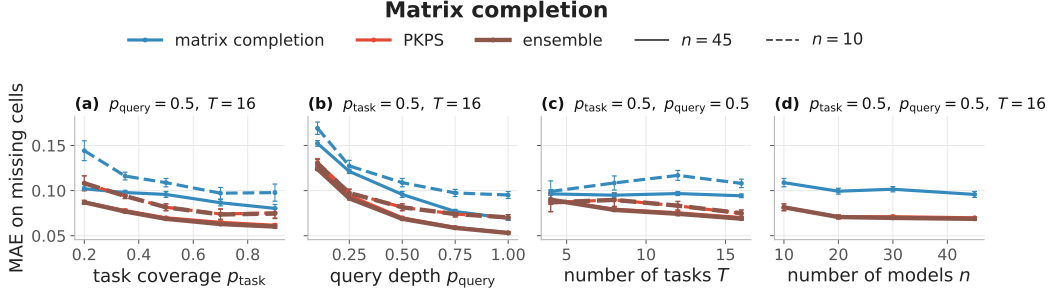


Figure 6: **Matrix completion on the Every Eval Ever suite** (MAE on missing cells,  $\pm 1$  SEM; solid  $n=45$ , dashed  $n=10$ ). Same protocol as Figure 4. Unlike the HELM suite, the embedding and ensemble beat low-rank completion at every operating point—including full cohort and coverage—consistent with the higher effective rank of this modern score matrix.

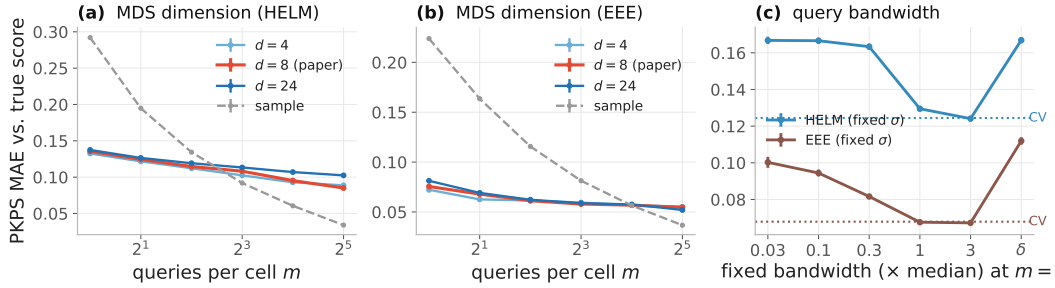


Figure 7: **Sensitivity analyses** on the query-efficiency experiment ( $\pm 1$  SEM, 16 seeds). (a, b) PKPS MAE vs. budget at MDS dimension  $d \in \{4, 8, 24\}$  (paper choice in red; raw sample dashed): the curves are nearly indistinguishable on both suites. (c) MAE at  $m=2$  with the query bandwidth *fixed* at each grid multiplier ( $\delta$  = exact matching); dotted lines mark the cross-validated selection, which matches the best fixed choice on both suites.

## D Proof of Theorem 1

**Population quantities.** Queries for model  $a$  are drawn i.i.d. from a design distribution over embedded queries; write  $z = (u, x_a(u))$  for an embedded query–response pair (response noise included) and  $P_a$  for its law. By Assumption 3 the kernels are bounded:  $k_Q \in [0, 1]$  after normalization (RBF) and  $|k_R| \leq B_R$ . Define the population affinity

$$A_{ab}^* = \frac{\mathbb{E}[k_Q(u, u') k_R(x_a(u), x_b(u'))]}{\mathbb{E}[k_Q(u, u')]}, \quad z \sim P_a, z' \sim P_b \text{ independent},$$

population squared distances  $D^{*2}(a, b) = A_{aa}^* + A_{bb}^* - 2A_{ab}^*$ , the centered Gram matrix  $B^* = -\frac{1}{2}JD^{*2}J$  with  $J = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ , and population perspectives  $\Psi^* \in \mathbb{R}^{n \times d}$  its rank- $d$  spectral embedding, which exists with  $\lambda_d(B^*) > 0$  by Assumption 3 (cf. Definition 1).

**Lemma 1** (Affinity concentration). *Under Assumption 3, for models  $a, b$  with  $\min(|Q_a|, |Q_b|) \geq m$  and any  $\delta \in (0, 1)$ : with probability at least  $1 - \delta$ ,*

$$|A_{ab} - A_{ab}^*| \leq \frac{2(1 + B_R)}{\kappa^2} t_m(\delta), \quad t_m(\delta) = \bar{B} \sqrt{\frac{\log(4/\delta)}{2m}}, \quad \bar{B} = \max(2B_R, 1),$$

provided  $t_m(\delta) \leq \kappa/2$ .

*Proof.* The numerator  $N = \frac{1}{|Q_a||Q_b|} \sum_{j,l} k_Q(u_j, u'_l) k_R(x_{aj}, x_{bl})$  and denominator  $Z = \frac{1}{|Q_a||Q_b|} \sum_{j,l} k_Q(u_j, u'_l)$  of Eq. 2 are two-sample  $U$ -statistics of order (1,1) in the i.i.d. pairs

$z_j \sim P_a, z'_j \sim P_b$ , with kernels of range at most  $2B_R$  and 1 respectively (response noise enters only through the bounded kernel argument, so the number of responses per query may stay fixed). By Hoeffding's inequality for two-sample  $U$ -statistics [Hoeffding, 1963, Section 5], each deviates from its mean by more than its range times  $\sqrt{\log(4/\delta)/2m}$  with probability at most  $\delta/2$ ; since  $\bar{B}$  dominates both ranges, a union bound gives  $|N - N^*| \leq t_m(\delta)$  and  $|Z - Z^*| \leq t_m(\delta)$  jointly with probability  $\geq 1 - \delta$ . On this event, since  $Z^* \geq \kappa$  by Assumption 3 and  $t_m(\delta) \leq \kappa/2$ , we have  $Z \geq \kappa/2$ , and

$$|A_{ab} - A_{ab}^*| = \frac{|NZ^* - N^*Z|}{ZZ^*} \leq \frac{|N - N^*|}{Z} + \frac{|N^*||Z - Z^*|}{ZZ^*} \leq \frac{2t_m(\delta)}{\kappa} + \frac{2B_R t_m(\delta)}{\kappa^2} \leq \frac{2(1+B_R)}{\kappa^2} t_m(\delta),$$

using  $|N^*| \leq B_R$  and  $\kappa \leq 1$ .

□

**Lemma 2** (Perspective perturbation). *Let  $\varepsilon_\infty = \max_{a,b} |A_{ab} - A_{ab}^*|$ . There is an orthogonal  $W \in \mathbb{R}^{d \times d}$  such that*

$$\max_i \|\hat{\psi}_i - W\psi_i\| \leq \frac{25\sqrt{d} n \varepsilon_\infty}{\sqrt{\lambda_d(B^*)}}.$$

*Proof.* Entrywise,  $|D^2 - D^{*2}| \leq 4\varepsilon_\infty$ , and double centering inflates an entrywise bound by at most a factor of 4, so  $E = B - B^*$  satisfies  $\|E\|_{\text{op}} \leq \|E\|_F \leq 8n\varepsilon_\infty$ . Let  $B_d$  be the top- $d$  truncation of  $B$  that classical MDS embeds ( $B_d = \hat{\Psi}\hat{\Psi}^\top$ , positive semidefinite). By Weyl's inequality every eigenvalue of  $B$  beyond the  $d$ -th, and every negative eigenvalue, has magnitude at most  $\|E\|_{\text{op}}$  (since  $\lambda_{d+1}(B^*) = 0$  and  $B^* \succeq 0$ ), so  $\|B_d - B\|_{\text{op}} \leq \|E\|_{\text{op}}$ . The matrix  $B_d - B^*$  has rank at most  $2d$ , hence

$$\|B_d - B^*\|_F \leq \sqrt{2d} \|B_d - B^*\|_{\text{op}} \leq \sqrt{2d} (\|B_d - B\|_{\text{op}} + \|E\|_{\text{op}}) \leq 2\sqrt{2d} \|E\|_F.$$

Both  $B_d = \hat{\Psi}\hat{\Psi}^\top$  and  $B^* = \Psi^*\Psi^{*\top}$  are positive semidefinite of rank at most  $d$ , so by the Procrustes alignment lemma of Tu et al. [2016, Lemma 5.4],

$$\min_{W \in O(d)} \|\hat{\Psi} - \Psi^*W\|_F^2 \leq \frac{\|B_d - B^*\|_F^2}{2(\sqrt{2}-1)\lambda_d(B^*)}.$$

Combining,  $\max_i \|\hat{\psi}_i - W\psi_i\| \leq \|\hat{\Psi} - \Psi^*W\|_F \leq \frac{2\sqrt{2d}}{\sqrt{2(\sqrt{2}-1)}} \cdot \frac{8n\varepsilon_\infty}{\sqrt{\lambda_d(B^*)}} \leq \frac{25\sqrt{d} n \varepsilon_\infty}{\sqrt{\lambda_d(B^*)}}$ . Nearest-neighbor distances are invariant to  $W$ . □

**Proof of Theorem 1.** Take scores in  $[0, 1]$  without loss of generality (Assumption 1). Fix  $\varepsilon > 0$  and set targets  $c \leq \sqrt{\varepsilon/2}/(8\gamma)$  for the estimation error and  $\varepsilon' \leq \sqrt{\varepsilon/2}/(2\gamma)$  for the population nearest-neighbor distance. By Assumption 2 there is  $n_0(\varepsilon')$  such that for  $n \geq n_0$ , a reference model within population distance  $\varepsilon'$  of the target exists except with probability at most  $\varepsilon/8$ . Fix such an  $n$ . Applying Lemma 1 with confidence  $\delta = \varepsilon/(8n^2)$  to every pair, a union bound and Lemma 2 give  $\max_i \|\hat{\psi}_i - W\psi_i\| \leq c$  except with probability at most  $\varepsilon/8$ , once

$$m \geq C' \frac{d(1+B_R)^2 \bar{B}^2 n^2 \log(n/\varepsilon)}{\kappa^4 \lambda_d c^2} = \text{poly}(n, 1/\kappa, 1/\varepsilon),$$

for an absolute constant  $C'$  (this choice also satisfies the proviso  $t_m \leq \kappa/2$  of Lemma 1), and does not involve the pairing fraction  $\rho$ . On the intersection of the two events, for a target model  $f$  with nearest estimated reference neighbor  $f^*$  (perspectives  $\psi^*, \hat{\psi}^*$ ; the target's  $\psi, \hat{\psi}$ ),

$$|\hat{y}(f) - y(f)| \leq \gamma \|\psi^* - \psi\| \leq \gamma (\|\hat{\psi}^* - W\psi^*\| + \|\hat{\psi}^* - \hat{\psi}\| + \|\hat{\psi} - W\psi\|) \leq \gamma(2c + \hat{\delta}),$$

using Assumption 1 and orthogonal invariance. The estimated nearest-neighbor distance  $\hat{\delta}$  is at most the estimated distance to the  $\varepsilon'$ -close reference, itself at most  $\varepsilon' + 2c$  by the same alignment argument, so  $|\hat{y} - y| \leq \gamma(4c + \varepsilon') \leq \sqrt{\varepsilon/2}$ . Squaring, and adding the two failure events (squared errors are bounded by 1),  $\text{MSE}(\hat{y}) \leq \varepsilon/2 + \varepsilon/4 \leq \varepsilon$ . □

**Contrast with paired DKPS.** With  $k_Q = \delta$ , only identically-shared query pairs enter the double sum, so the  $U$ -statistic runs over the  $\rho M_{ij}$  shared queries and the rate in Lemma 1 becomes  $O_P((\rho M_{ij})^{-1/2})$ : DKPS’s guarantee requires  $\rho M_{ij} \rightarrow \infty$  and vacates in the unpaired limit  $\rho \rightarrow 0$ , where its distance is undefined (Section 3). This asymmetry motivates Conjecture 1; a formal lower bound for DKPS is future work.